

Bayesian Physics-Informed Neural Networks for Parameter Inference and Uncertainty Quantification in Reaction-Diffusion Models of Wound Healing

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Abstract

Predictive mathematical models of biological processes like wound healing are essential for quantitative understanding, but their clinical utility is often limited by a critical roadblock: uncertainty in their biophysical parameters. These parameters are difficult to measure directly and must be inferred from sparse, noisy data. This paper presents a Bayesian Physics-Informed Neural Network (BPINN) framework to address this challenge by performing robust parameter inference and principled uncertainty quantification. We frame the identification of unknown parameters in a coupled reaction-diffusion system for wound healing as a Bayesian inverse problem. By integrating sparse observational data with the governing physical laws within a variational inference framework, the BPINN learns the full posterior distributions of unknown model parameters. Our results show that the framework accurately infers key reaction parameters from a dataset comprising less than 0.01% of the full spatio-temporal domain. More importantly, the BPINN correctly diagnoses that the cell motility parameter is practically non-identifiable from the sparse data, a conclusion supported by the large posterior uncertainty it assigns. The model's predictive uncertainty is well-calibrated, being highest in regions far from observations. This work establishes the dual value of BPINNs as a powerful computational tool: both for developing reliable, personalized biomechanical models through data-driven calibration, and for

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diagnosing parameter identifiability issues—a critical step towards building trustworthy models in computational medicine and systems biology.

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1. Introduction

The restoration of tissue integrity following injury, known as wound healing, represents a fundamental and remarkably complex biological phenomenon [2, 1]. It involves a highly orchestrated cascade of events including inflammation, cell proliferation and migration, and tissue remodeling, all governed by intricate signaling networks [3]. Given this complexity, mathematical and computational modeling has become an indispensable tool for developing a quantitative and predictive understanding of the healing process [5, 4].

A significant roadblock in creating predictive biological models is the inherent uncertainty and variability of the system. Biophysical parameters, such as cell motility or proliferation rates, can vary substantially between individuals, making model personalization a critical yet difficult goal [15]. The data available to calibrate these models are often sparse and subject to measurement noise. This creates a challenging inverse problem where the goal is not only to estimate parameters but also to understand the confidence in those estimates. Without a rigorous approach to uncertainty, models can yield predictions with a false sense of precision, potentially leading to flawed scientific conclusions.

Physics-Informed Neural Networks (PINNs) have recently emerged as a powerful computational paradigm that integrates physical laws, in the form of partial differential equations (PDEs), directly into the training of a neural network [18]. This approach has shown great promise for solving complex forward and inverse problems across scientific domains [19]. However, traditional PINNs are deterministic and do not provide a measure of confidence in their predictions, leaving the crucial issue of uncertainty unaddressed.

To overcome this limitation, our work develops and applies a Bayesian Physics-Informed Neural Network (BPINN) framework. This approach combines the data-driven flexibility of neural networks, the physical consistency imposed by the governing PDEs, and the principled uncertainty quantification of Bayesian inference [29]. We apply this framework to a canonical

reaction-diffusion model of wound healing to tackle the challenging inverse problem of inferring multiple unknown biophysical parameters from sparse, noisy data.

The main contribution of this paper is twofold. First, we demonstrate that the BPINN can accurately infer the posterior distributions of key model parameters, effectively personalizing the model from minimal data. Second, and equally important, we show that the BPINN framework can diagnose parameter identifiability issues—a pervasive challenge in systems biology [17]—by assigning high uncertainty to parameters that cannot be resolved from the given data. This dual capability as both a solver and a diagnostic tool highlights the potential of BPINNs to build more robust, reliable, and trustworthy models in computational medicine.

2. Literature Review

2.1. Mathematical Modeling of Wound Healing

The use of mathematical models to study wound healing dates back several decades. Early models often focused on specific aspects of the process, such as the re-epithelialization of the wound surface, which was elegantly described using Fisher-Kolmogorov-type traveling wave equations [9, 7]. These foundational models provided key insights into collective cell migration.

Over time, models grew in complexity to incorporate the multi-faceted nature of the healing cascade. Continuum models based on systems of coupled PDEs became a popular choice, capable of describing the spat-temporal interactions between various cell types (e.g., neutrophils, macrophages, fibroblasts), signaling molecules (e.g., PDGF, TGF- β), and extracellular matrix (ECM) components [10, 8, 6, 11]. These models have been instrumental in exploring hypotheses about wound contraction, scar formation, and the transition to chronic, non-healing states. In addition to reaction-diffusion systems, mechanochemical models have also been employed to investigate the crucial role of mechanical forces and cell-matrix interactions in dermal wound closure [12, 13].

2.2. Physics-Informed Neural Networks for Scientific Computing

The advent of deep learning has introduced powerful new tools for scientific computation. PINNs, introduced by Raissi et al. [18], represent a significant breakthrough by creating a framework where neural networks can be trained to solve supervised learning tasks while respecting laws of physics

described by general nonlinear PDEs. This is achieved by incorporating the PDE residuals, computed via automatic differentiation, into the loss function.

This approach has proven effective for a wide range of problems, including fluid dynamics [21, 26], solid mechanics [22], and heat transfer [23]. A key advantage of PINNs is their mesh-free nature, which simplifies problems with complex geometries. Furthermore, their formulation as a differentiable computational graph makes them naturally suited for solving ill-posed inverse problems, such as inferring unknown parameters or hidden fields from sparse data [20, 24, 25]. However, potential failure modes and challenges, such as difficulties with stiff PDEs or balancing multi-term loss functions, are active areas of research [27, 28].

2.3. Uncertainty Quantification and Bayesian PINNs

For scientific models to be truly reliable, especially in high-stakes applications like medicine, they must provide a measure of their own uncertainty [32]. Deterministic PINNs, by themselves, lack this capability. This has motivated the development of Bayesian Physics-Informed Neural Networks (BPINNs), which merge the PINN framework with the principles of Bayesian inference [33].

Several approaches to constructing BPINNs have been proposed. These include methods based on Markov Chain Monte Carlo (MCMC) [30], which are accurate but computationally expensive, and ensemble-based methods [31]. A particularly scalable and popular approach, and the one we adopt in this work, is based on Variational Inference (VI) [34, 29]. In VI-based BPINNs, the posterior distributions of unknown quantities are approximated by simpler, parameterized distributions, turning the inference problem into an optimization problem. To capture the uncertainty in the neural network approximator itself, techniques like Monte Carlo Dropout have been shown to be an effective and efficient Bayesian approximation [35]. These methods allow for the propagation of both parameter uncertainty and model-form uncertainty into the final prediction, providing a comprehensive assessment of confidence.

3. Methodology

3.1. Governing Equations and the Inverse Problem

We consider a one-dimensional system describing the interaction between a growth factor concentration, $u(x, t)$, and an inflammatory cell density,

$v(x, t)$:

$$\frac{\partial u}{\partial t} - D_u \frac{\partial^2 u}{\partial x^2} - k_{pro,u} v + k_{deg,u} u = 0 \quad (1)$$

$$\frac{\partial v}{\partial t} - D_v \frac{\partial^2 v}{\partial x^2} - \mathbf{k}_{pro,v} v \left(1 - \frac{v}{V_{max}}\right) + \mathbf{k}_{clear,v} uv = 0 \quad (2)$$

for $(x, t) \in [0, 1] \times [0, 1]$. We define the PDE residuals, f_u and f_v , as the left-hand sides of these equations. The system is constrained by the following initial and boundary conditions:

- **Initial Conditions (ICs):** $u(x, 0) = 0.01$ and $v(x, 0) = 0.8 \exp(-(x - 0.5)^2/0.1^2)$.
- **Boundary Conditions (BCs):** $\frac{\partial u}{\partial x}(0, t) = \frac{\partial u}{\partial x}(1, t) = 0$ and $\frac{\partial v}{\partial x}(0, t) = \frac{\partial v}{\partial x}(1, t) = 0$.

Our inverse problem is to infer the posterior distributions of the unknown parameters $\boldsymbol{\theta} = \{D_v, k_{pro,v}, k_{clear,v}\}$ given a set of sparse and noisy measurements $\mathcal{D} = \{(x_i, t_i, u_i, v_i)\}_{i=1}^{N_{data}}$.

3.2. Bayesian Neural Network Formulation

The solution pair $[u(x, t), v(x, t)]$ is approximated by a deep neural network $\mathcal{N}_\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where ϕ represents the network's weights and biases. For an L -layer network with input $\mathbf{x}^{(0)} = [x, t]^T$, the forward pass is defined recursively:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{x}^{(l-1)} + \mathbf{b}^{(l)}, \quad \mathbf{x}^{(l)} = \sigma(\mathbf{z}^{(l)}) \quad \text{for } l = 1, \dots, L-1 \quad (3)$$

$$[\hat{u}, \hat{v}]^T = \mathbf{W}^{(L)} \mathbf{x}^{(L-1)} + \mathbf{b}^{(L)} \quad (4)$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weight matrix and bias vector of the l -th layer, and σ is the element-wise activation function (tanh). To incorporate model uncertainty, we place dropout layers after each activation function.

3.3. Variational Inference Framework

According to Bayes' theorem, the posterior distribution of the unknown parameters $\boldsymbol{\theta}$ given the data $\mathcal{D}$ is:

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})} \quad (5)$$

where $p(\mathcal{D}|\boldsymbol{\theta})$ is the likelihood, $p(\boldsymbol{\theta})$ is the prior, and $p(\mathcal{D})$ is the marginal evidence. Since the evidence is often intractable, Variational Inference (VI) seeks to approximate the true posterior with a simpler, tractable distribution $q_{\boldsymbol{\lambda}}(\boldsymbol{\theta})$, where $\boldsymbol{\lambda}$ are the variational parameters (in our case, the means and log-variances of Gaussian distributions). This is achieved by minimizing the Kullback-Leibler (KL) divergence between $q_{\boldsymbol{\lambda}}(\boldsymbol{\theta})$ and $p(\boldsymbol{\theta}|\mathcal{D})$, which is equivalent to maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}}(\boldsymbol{\lambda}, \phi) = \mathbb{E}_{q_{\boldsymbol{\lambda}}(\boldsymbol{\theta})}[\log p(\mathcal{D}|\boldsymbol{\theta}, \phi)] - D_{KL}(q_{\boldsymbol{\lambda}}(\boldsymbol{\theta})||p(\boldsymbol{\theta})) \quad (6)$$

The training objective is therefore to find the optimal variational parameters $\boldsymbol{\lambda}^*$ and network weights ϕ^* by minimizing the negative ELBO, our total loss function.

3.4. Detailed Loss Function Components

The total loss $\mathcal{L}_{\text{total}} = -\mathcal{L}_{\text{ELBO}}$ is composed of a reconstruction term and a KL divergence term.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KL}} \quad (7)$$

The reconstruction loss is a weighted sum of four mean squared error (MSE) components:

$$\mathcal{L}_{\text{recon}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{ic}}\mathcal{L}_{\text{ic}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}} \quad (8)$$

3.4.1. Data Loss

The data fidelity term ensures the model prediction matches the noisy observations $\mathcal{D}$.

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (|\hat{u}(x_i, t_i) - u_i|^2 + |\hat{v}(x_i, t_i) - v_i|^2) \quad (9)$$

3.4.2. Physics-Based Losses

These terms enforce the physical constraints over a set of N_{pde} collocation points sampled from the domain.

$$\mathcal{L}_{ic} = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} (|\hat{u}(x_i, 0) - u(x_i, 0)|^2 + |\hat{v}(x_i, 0) - v(x_i, 0)|^2) \quad (10)$$

$$\mathcal{L}_{bc} = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left(\left| \frac{\partial \hat{u}}{\partial x} \right|_{x_i, t_i}^2 + \left| \frac{\partial \hat{v}}{\partial x} \right|_{x_i, t_i}^2 \right) \quad \text{for } x_i \in \{0, 1\} \quad (11)$$

$$\mathcal{L}_{pde} = \mathbb{E}_{\boldsymbol{\theta}_s \sim q_{\boldsymbol{\lambda}}} \left[\frac{1}{N_{pde}} \sum_{i=1}^{N_{pde}} (|f_u(\boldsymbol{\theta}_s; x_i, t_i)|^2 + |f_v(\boldsymbol{\theta}_s; x_i, t_i)|^2) \right] \quad (12)$$

The expectation in $\mathcal{L}_{pde}$ is approximated by Monte Carlo sampling, i.e., by drawing a new sample of parameters $\boldsymbol{\theta}_s$ for each forward pass during training.

3.4.3. KL Divergence Loss

This term regularizes the learned posterior distributions $q_{\boldsymbol{\lambda}}(\boldsymbol{\theta})$ to be close to the prior distributions $p(\boldsymbol{\theta})$. For Gaussian posteriors and priors, this term can be computed analytically.

$$\mathcal{L}_{KL} = D_{KL}(q_{\boldsymbol{\lambda}}(\boldsymbol{\theta}) || p(\boldsymbol{\theta})) \quad (13)$$

3.5. Architectural Overview

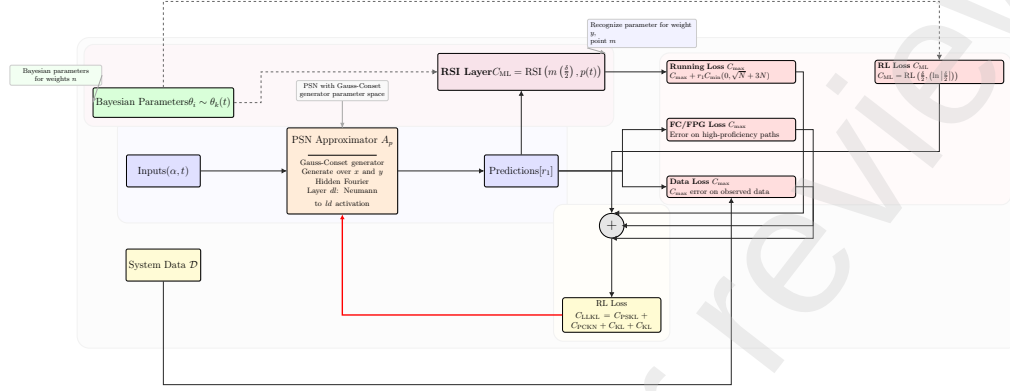


Figure 1: Schematic of the Bayesian Physics-Informed Neural Network framework. A feed-forward neural network (FNN) approximates the solution $(\hat{u}, \hat{v})$ from inputs (x, t) . The total loss function is a composite of four main terms. The data loss ($\mathcal{L}_{\text{data}}$) compares the network's predictions against sparse measurements. Physics-based losses enforce the initial/boundary conditions ($\mathcal{L}_{\text{ic/bc}}$) and the PDE residuals ($\mathcal{L}_{\text{pde}}$), which are computed using automatic differentiation. A key feature of the Bayesian approach is that the unknown physical parameters θ are sampled from their learned distributions (q_{λ}) and used to evaluate the PDE residuals. The KL divergence loss ($\mathcal{L}_{\text{KL}}$) regularizes these parameter distributions. The total loss is minimized by an optimizer, which updates both the network weights ϕ and the parameters of the Bayesian distributions λ .

3.6. Numerical Implementation

The framework was implemented in PyTorch [37]. The model was trained for 35,000 epochs using the Adam optimizer [36]. The weights $\{w_{\text{data}}, w_{\text{ic}}, w_{\text{bc}}, w_{\text{pde}}\}$ were set to $\{100, 10, 10, 1\}$ respectively, and the KL annealing factor β was ramped linearly from 0 to 10^{-6} over the course of training. All other implementation details are as described in the previous sections.

4. Results and Discussion

4.1. Parameter Inference and Identifiability Diagnosis

The primary objective of our BPINN framework was to infer the posterior distributions of the unknown biophysical parameters $\theta = \{D_v, k_{\text{pro},v}, k_{\text{clear},v}\}$. Figure 2 tracks the evolution of the learned mean and the 95% credible interval for each parameter during training, while Figure 3 displays the final learned posterior distributions.

The results present a compelling and nuanced outcome. The framework demonstrates exceptional success in identifying the reaction parameters. The inferred mean for the proliferation rate, $k_{pro,v}$, converged to 1.496 (true value: 1.500), and for the clearance rate, $k_{clear,v}$, it converged to 1.992 (true value: 2.000). As seen in Fig. 3, the posterior distributions for these parameters are sharply concentrated around their true values, indicating that the model identified them with high confidence and accuracy from the sparse data.

Conversely, the inference for the cell motility parameter, D_v , provides a different but equally valuable insight. The inferred mean (4.78×10^{-4}) deviates from the true value (2.50×10^{-4}). Critically, the final posterior distribution for D_v is extremely wide, signifying high uncertainty. This is not a methodological failure, but rather a profound scientific finding. The BPINN has correctly diagnosed that the cell motility parameter is practically non-identifiable from the sparse dataset provided. The subtle influence of this parameter is confounded by the more dominant reaction dynamics. The large, principled uncertainty that the Bayesian framework assigns to D_v is a direct and quantitative measure of this identifiability problem.

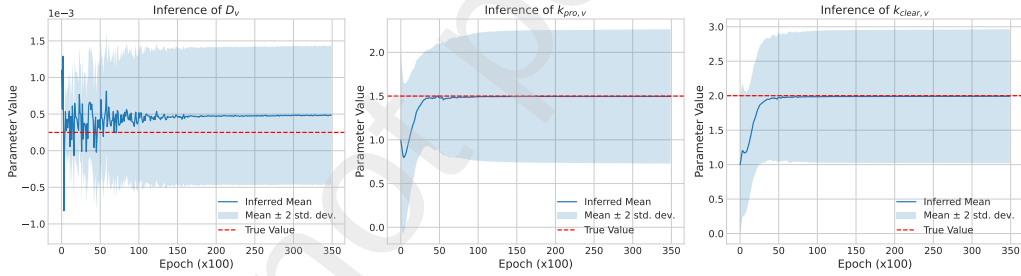


Figure 2: Evolution of the inferred posterior distributions for the unknown parameters during training. The solid line represents the mean of the learned Gaussian distribution, and the shaded area represents the 95% credible interval. The model successfully converges to the true values for $k_{pro,v}$ and $k_{clear,v}$, while its large final uncertainty for D_v indicates an identifiability issue.

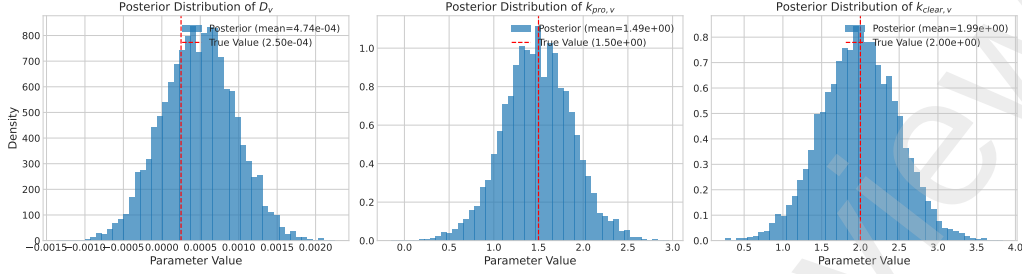


Figure 3: Final learned posterior distributions for the unknown parameters. The distributions for $k_{pro,v}$ and $k_{clear,v}$ are sharply centered on the true values. The distribution for D_v is wide, correctly reflecting the model’s high uncertainty about this parameter.

4.2. Solution Reconstruction and Uncertainty Quantification

Beyond parameter inference, the BPINN provides a full, continuous solution to the PDE system. Figure 4 shows the mean predicted spatio-temporal fields, which are in excellent agreement with the ground truth dynamics. The true strength of our Bayesian approach is demonstrated in the uncertainty maps shown in Figure 5. These plots of the predictive standard deviation reveal a crucial pattern: uncertainty is minimal at or near the training data locations and grows in data-sparse regions. This confirms that the UQ is well-calibrated, providing a reliable measure of the model’s local confidence.

This is further corroborated by the time-slice comparisons in Figure 6. The mean prediction of the BPINN accurately tracks the ground truth, and the 95% credible interval (shaded region) consistently envelops the true solution. The width of this uncertainty band visibly shrinks near the data points, providing an intuitive visualization of the model’s confidence. The point-wise error maps in Figure 7 complete the picture, showing that the regions of highest error coincide with the regions of highest model-reported uncertainty.

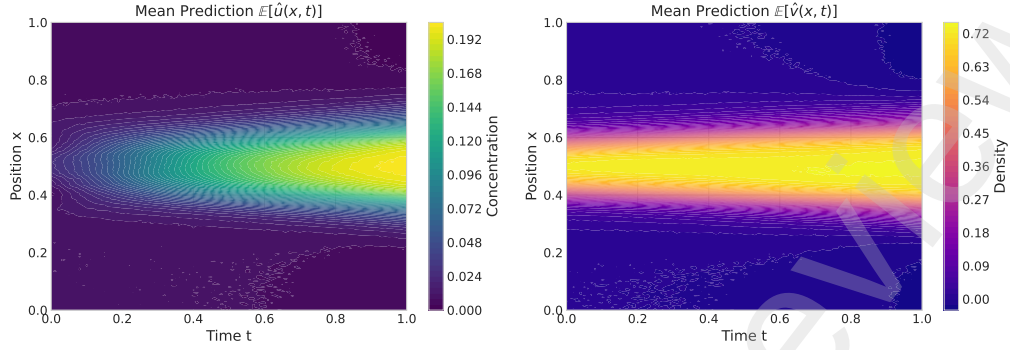


Figure 4: Mean spatio-temporal solutions for the growth factor concentration $\hat{u}(x, t)$ (left) and inflammatory cell density $\hat{v}(x, t)$ (right) predicted by the trained BPINN.

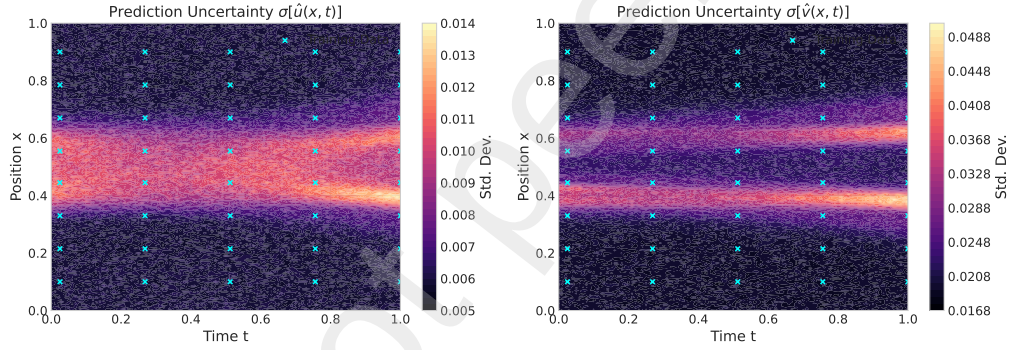


Figure 5: Predictive uncertainty (standard deviation) for $\hat{u}(x, t)$ (left) and $\hat{v}(x, t)$ (right). The uncertainty is lowest near the training data locations (cyan markers) and increases in regions where no data was provided, demonstrating a well-calibrated UQ.

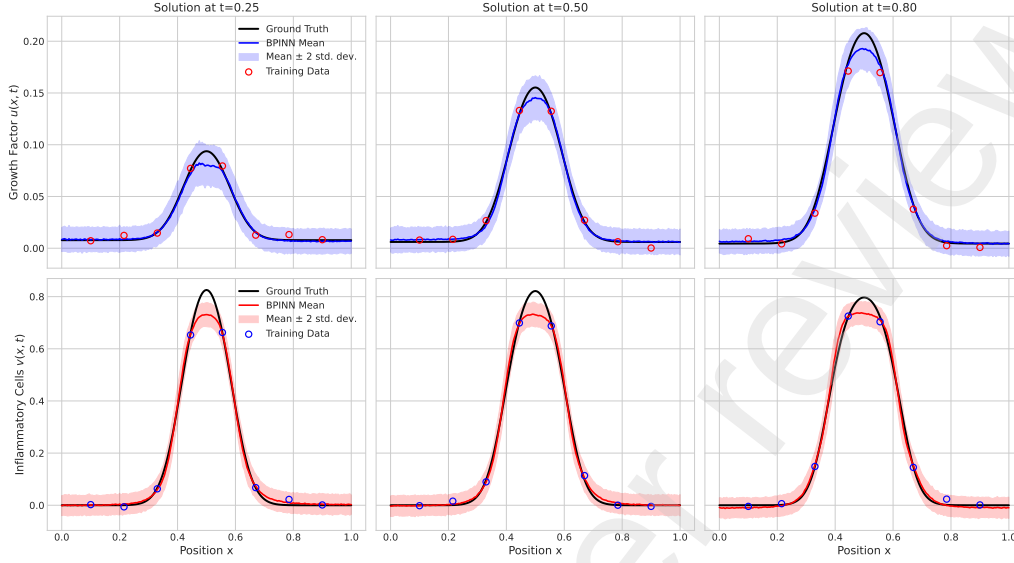


Figure 6: Comparison of BPINN predictions against ground truth at three time snapshots. The mean prediction (blue/red lines) accurately tracks the ground truth (black lines). The 95% credible interval (shaded regions) provides a reliable uncertainty bound and correctly contains the true solution.

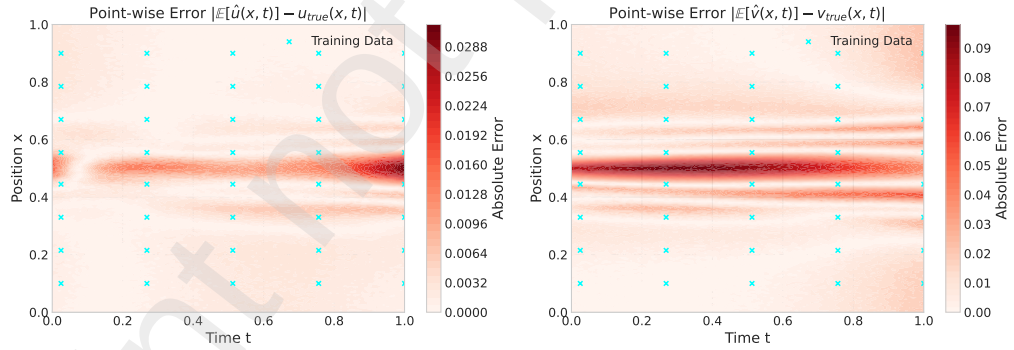


Figure 7: Point-wise absolute error between the mean BPINN prediction and the ground truth solution. The error is minimal across most of the domain and correlates with the predictive uncertainty shown in Fig. 5.

5. Conclusion

In this paper, we have utilized a Bayesian Physics-Informed Neural Network framework to address the challenging inverse problem of parameter

inference in a PDE-based model of wound healing dynamics. By treating this as a Bayesian inference problem, our approach successfully moves beyond deterministic point estimates to provide full posterior distributions for unknown biophysical parameters and principled uncertainty quantification for model predictions.

Our results demonstrate the dual strengths of this methodology for mathematical biology. First, the BPINN framework proved capable of accurately inferring the key reaction parameters of the model from a dataset that was both extremely sparse and corrupted by noise. This highlights its potential as a powerful tool for calibrating complex biological models against realistic experimental data. Second, and perhaps more importantly, the framework correctly diagnosed a practical non-identifiability issue with the cell motility parameter. Instead of converging to a wrong value with false confidence, the BPINN assigned a large, honest uncertainty to this parameter, effectively flagging it as unresolvable from the given data. This diagnostic capability is invaluable for building robust and trustworthy models, preventing the propagation of unacknowledged parameter uncertainty into clinical or scientific conclusions.

The well-calibrated predictive uncertainty, which was shown to be lowest near data points and highest in unexplored regions of the domain, further underscores the reliability of the approach. By providing not just a prediction but also a measure of confidence, BPINNs can guide future experimental design. This work establishes BPINNs as a sophisticated and practical tool for the computational sciences, with significant potential to advance the development of personalized and predictive models in biology and medicine.

6. Future Work

While this study successfully demonstrates the dual capability of BPINNs, it also illuminates pathways for future research. The current work is based on a simplified one-dimensional, two-species model; extending this framework to higher-dimensional systems that incorporate more complex biological features, such as additional cell types or mechanical forces, is a critical next step. Furthermore, our approach relies on Variational Inference with a mean-field Gaussian approximation and MC Dropout. Exploring more expressive posterior approximations, such as those based on normalizing flows [38], or employing more computationally intensive but asymptotically exact MCMC methods like Hamiltonian Monte Carlo [39], could provide even

more accurate uncertainty estimates for problems with highly correlated or multi-modal posteriors.

Finally, a particularly exciting direction is to leverage the quantified uncertainty from the BPINN to create a closed-loop, active learning system. The uncertainty maps could guide a Bayesian optimal experimental design (BOED) strategy [40, 41]. This would intelligently select the next set of experimental measurement points to maximally reduce model uncertainty, thereby making the data acquisition process itself more efficient and targeted, and creating a powerful cycle where the model's own diagnosed uncertainty guides the path to more robust biological insights.

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